Plasmid preparation by alkaline lysis

Risk assessment

▷ Wear gloves, safety glasses, lab coat

Reviewed: May 21, 2026

This is a bench card. Full protocol available online.

›› *Alkaline lysis*

(1.) Grow bacteria in 1–5 mL LB (Miller) overnight to stationary phase. Collect the cells by centrifugation.

(2.) Resuspend the pellet in 150 µL Buffer P1.

(3.) Add 250 µL Buffer P2. Mix by gently inverting the tube 6–8 times. Incubate at room temperature for no more than 5 min or until the lysate appears clear.

> ***Critical:*** Do not vortex or pipette to avoid shearing chromosomal DNA.

(4.) Add 350 µL Buffer N3. Mix gently by inversion 6–8 times.

> ***Critical:*** Ensure complete neutralization to precipitate protein and chromosomal DNA. If Buffer P2 contains thymolphthalein, mix until the solution is colorless.

(5.) Centrifuge at 17 000 × *g* for 3 min to pellet precipitated material. Repeat if the lysate remains turbid.

(6.) Transfer the clear supernatant to a fresh tube for purification. Avoid any carryover of precipitate.

A › *Purification of plasmid DNA over a silica matrix*

(1.) Apply up to 700 µL of supernatant to a silica spin column. Centrifuge at 11 000 × *g* for 1 min, or use a vacuum manifold.

(2.) *Optional:* Wash the silica matrix with 450 µL Buffer PB.

(3.) Wash with 650 µL Buffer PE.

(4.) Dry the silica matrix by centrifugation at 11 000 × *g* for 2 min.

(5.) Place the spin column into a clean microcentrifuge tube.

(6.) Apply 25–55 µL 5 mM Tris pH 8.0 to the membrane. Incubate for 1 min, then elute the plasmid by centrifugation at 11 000 × *g* for 1 min.

(7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

B › *Purification by alcohol precipitation*

(1.) Transfer up to 700 µL of supernatant to a clean microcentrifuge tube.

(2.) Add an equal volume of isopropyl alcohol, or 2 vol of ethanol, to precipitate the DNA. Vortex to mix thoroughly.

(3.) Centrifuge at 11 000 × *g* for 5 min. Discard the supernatant and briefly spin to remove residual solvent. The pellet will appear nearly transparent.

(4.) Wash with 650 µL Buffer PE or freshly prepared 70% ethanol without disturbing the pellet.

(5.) Air-dry the pellet for 2–5 min, or dry in a vacuum centrifuge for 1 min.

(6.) Resuspend in 25–55 µL 5 mM Tris pH 8.0 or water.

C › *Purification of plasmid DNA by anion-exchange chromatography*

(1.) Equilibrate the anion-exchange column with the recommended volume of Buffer QBT and allow it to drain by gravity flow.

(2.) Apply the P3-neutralized, cleared lysate to the equilibrated column and let it pass through by gravity.

> ***Critical:*** The column must not run dry between steps; if flow stalls, refill with Buffer QC and continue.

(3.) Wash the column twice with the recommended volume of Buffer QC.

(4.) Elute the plasmid DNA with the recommended volume of Buffer QF into a clean tube.

(5.) Precipitate the DNA by adding 0.7 vol isopropanol at room temperature, wash, and resuspend as described in *Alternative B*.

Recipe (available online) Troubleshooting (available online) Notes (available online)

Lab Protocols — © B. C. Buchmuller. Licensed CC-BY-SA-4.0. Research use; provided as-is, without warranty. Current when printed. Visit https://benjbuch.github.io/check/?q=3361393 or scan the QR code to read the full protocol or to check for updates.